## Questions

**Q1.**

Figure 7 shows a person trying to lift a large rock using a metal bar.

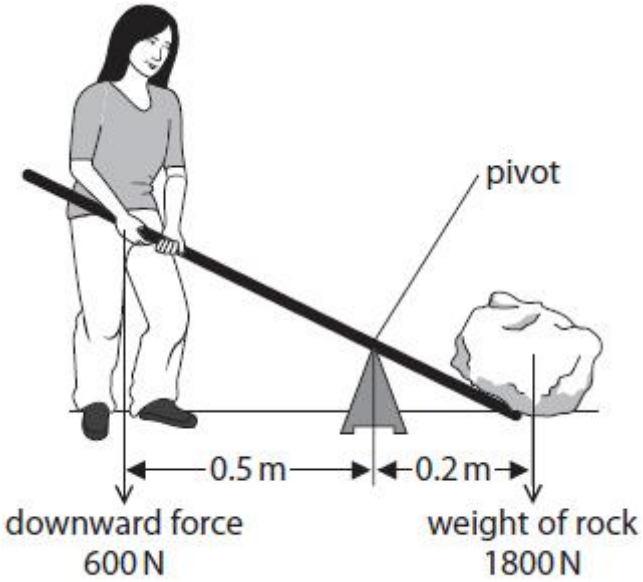


**Figure 7**

The rock weighs 1800 N.

The person can only produce a downwards force of 600 N.

The person cannot lift the rock.

(i) Explain, using calculations, why the person cannot lift the rock.

**(3)**

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(ii) Explain **one** change to the arrangement that will make it possible for this person to lift the rock.

**(2)**

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**(Total for question = 5 marks)**

**Q2.**

Figure 2 shows an open door.

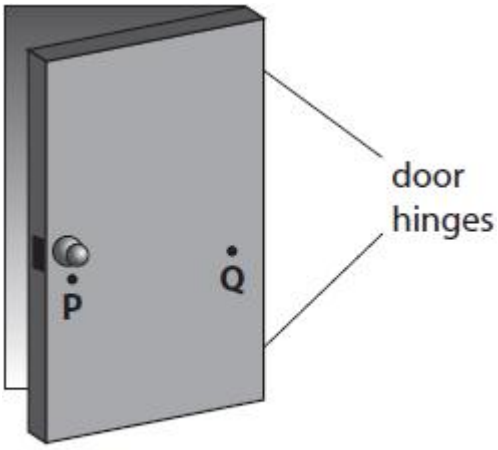


**Figure 2**

Explain why it is easier to close the door by pushing at point **P** rather than pushing at point **Q**.

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**(Total for question = 2 marks)**

**Q3.**

Figure 12 shows some of the bones and muscles in an arm.

The arrows show the forces on the forearm when the arm is bent.

The hand is empty.

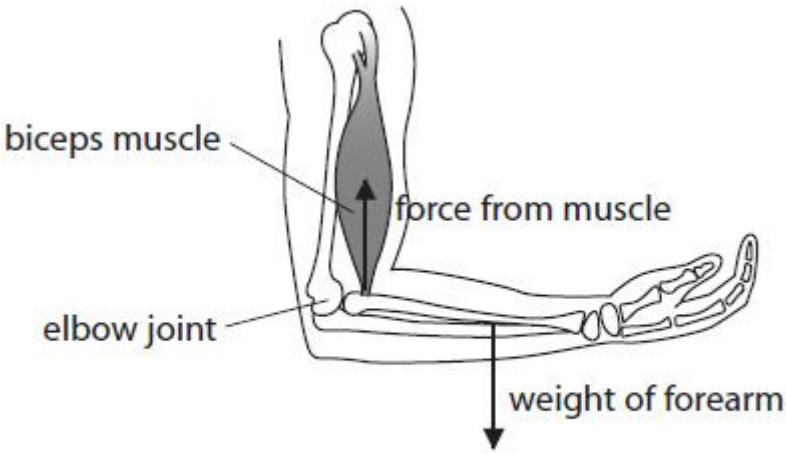


**Figure 12**

The biceps muscle provides a force to balance the weight of the forearm.

The weight of the forearm can be represented as a single force.

Figure 13 shows a diagram representing the forces and distances involved.

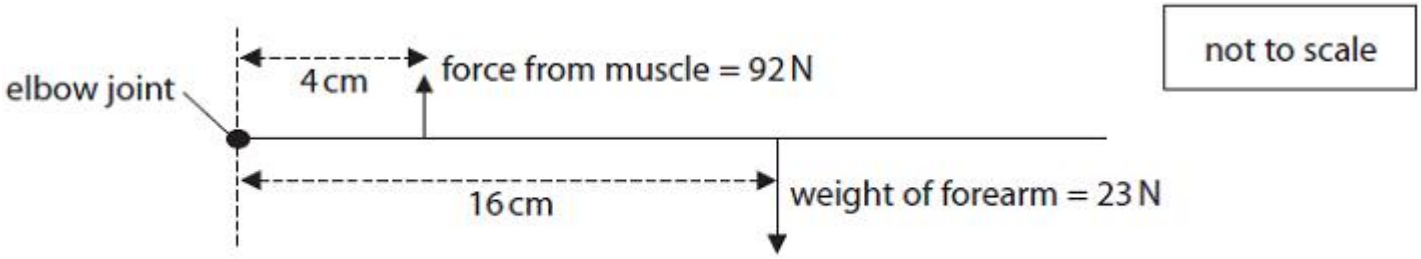


**Figure 13**

(i) Use the principle of moments to show that the system shown in Figure 13 is in equilibrium.

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(ii) The person then holds a ball weighing 15 N in their hand.

Figure 14 shows the forces on the forearm and their distances from the elbow joint.

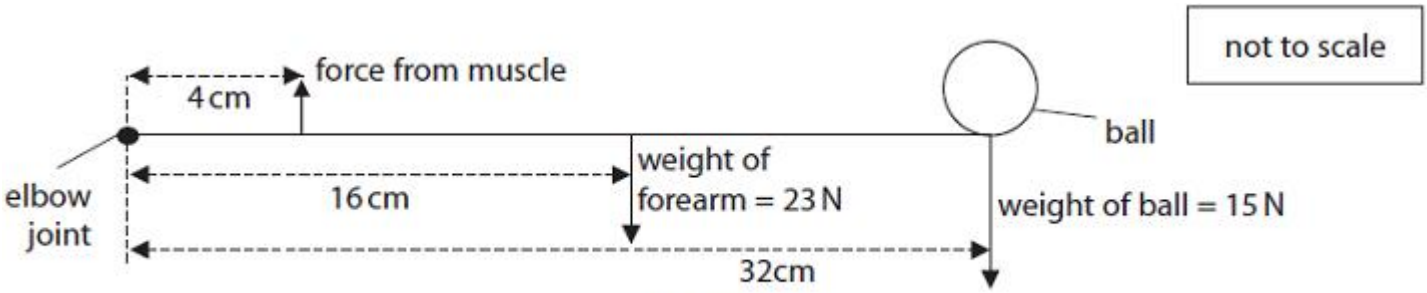


**Figure 14**

Calculate the force from the muscle that is needed to keep the system in Figure 14 in equilibrium.

(3)

force = ..... N

**(Total for question = 5 marks)**

**Q4.**

*Figure 24 shows a wooden block connected to a weight by a string.
The string goes over a pulley.

The surfaces of the table and the wooden block are both rough.

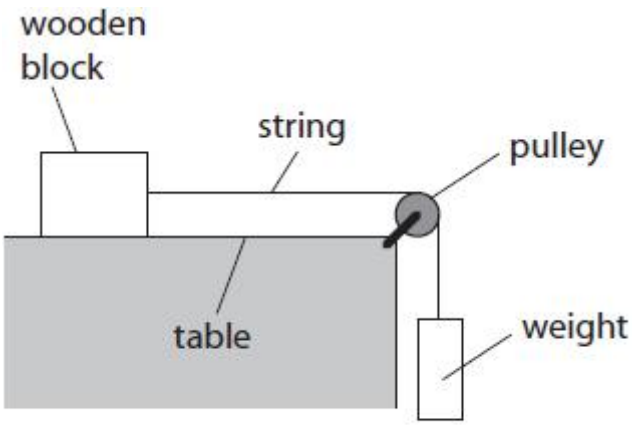


**Figure 24**

The wooden block moves across the table at a constant horizontal velocity.

Several vertical and horizontal forces act on the wooden block as it moves.

Explain how the forces keep the wooden block moving across the table at a constant horizontal velocity.

Your answer should refer to all forces acting on the wooden block.

You may add to the diagram to help with your answer.

(6)

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**(Total for question = 6 marks)**

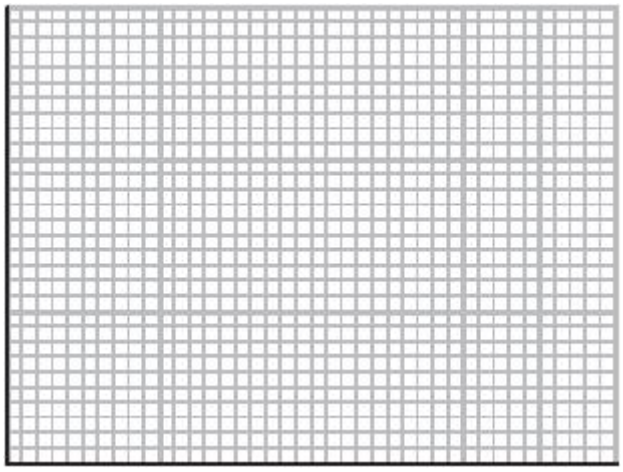
**Q5.**

Figure 7 shows two astronauts in space pushing at a satellite.

**Figure 7**

(i) The force $F_1$ is 3.0 N and the force $F_2$ is 2.0 N, acting at right angles to each other.
Draw a vector diagram to scale showing these forces.

**(2)**



(ii) Use the diagram in (i) to estimate the magnitude of the resultant force acting on the satellite.

**(2)**

resultant force = ..... N

**(Total for question = 4 marks)**

**Q6.**

Figure 23 shows two small boats pulling a much larger ship.

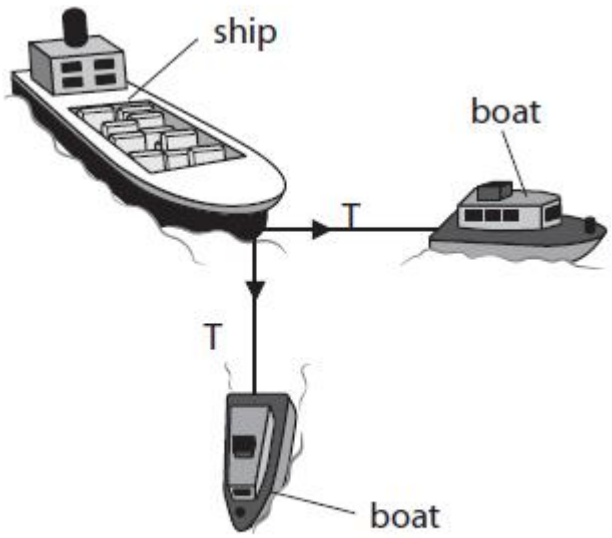



**Figure 23**

The ship is connected to the boats with ropes.

The tension, T, in each of the ropes has a magnitude of 20 kN.

The ropes are at right angles to each other.

Draw a vector diagram and use it to determine the resultant force that the boats exert on the ship.

**(4)**

magnitude of resultant force on the ship = ..... kN

**(Total for question = 4 marks)**

**Q7.**

Figure 7 shows a drone.

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**Figure 7**

The drone has four spinning blades.

The upward force produced enables the drone to rise in the air.

The speed at which the blades spin is measured in turns per minute.

Figure 8 shows how the upward force produced by the four blades depends on the speed at which the blades spin.

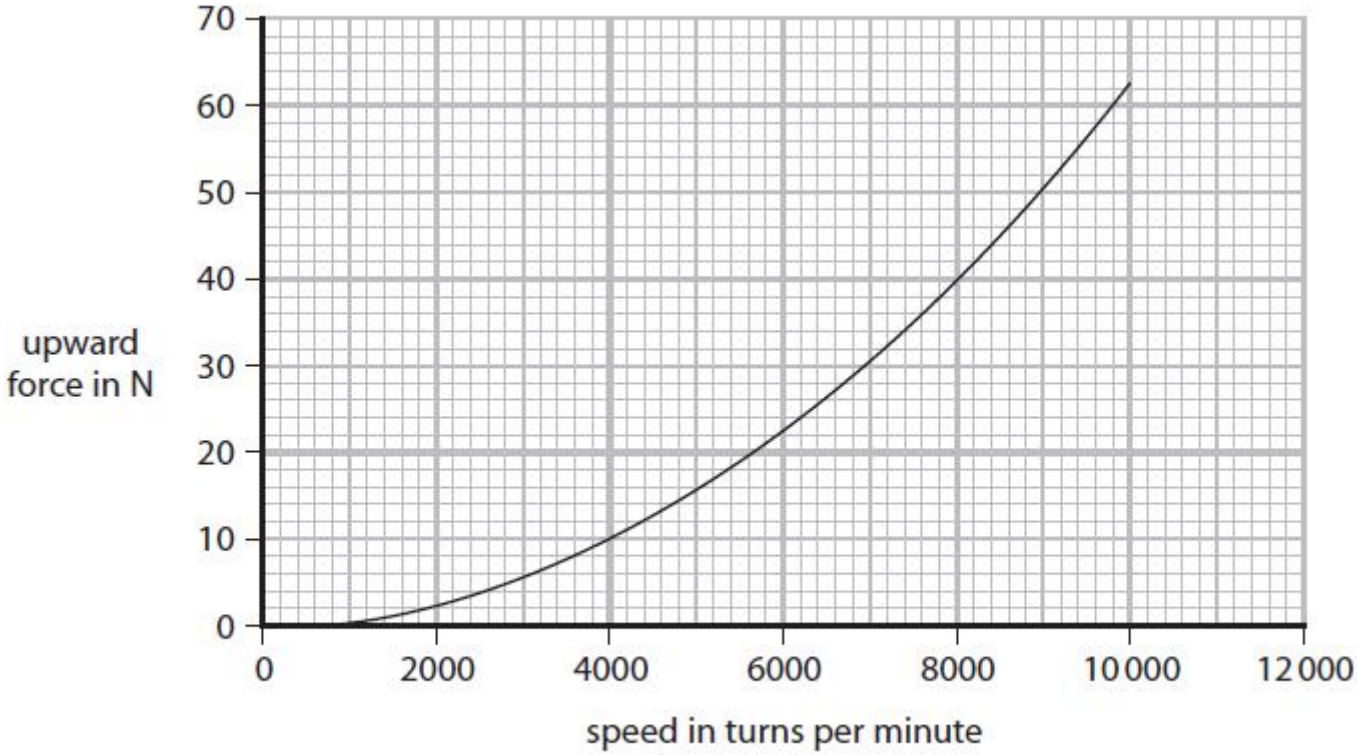


**Figure 8**

Describe the relationship between upward force and speed shown by this graph.

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**(Total for question = 2 marks)**

**Q8.**

Figure 9 shows a toy that has a plastic cylinder, a plastic base and two similar magnets. Each of the two magnets is in the shape of a ring.

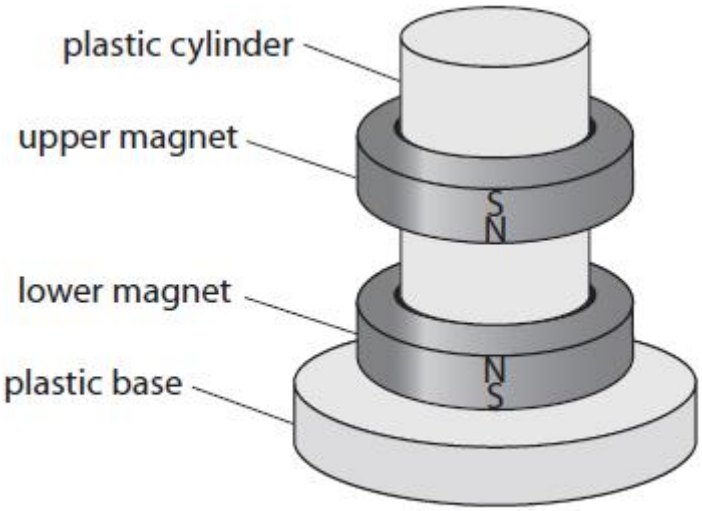


**Figure 9**

The upper magnet seems to float in the air above the lower magnet.

Describe the forces acting on the upper magnet.

Use the idea of magnetic fields in your answer.

**(3)**

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**(Total for question = 3 marks)**

**Q9.**

A student investigates moments of forces.

Figure 14 shows the apparatus used.

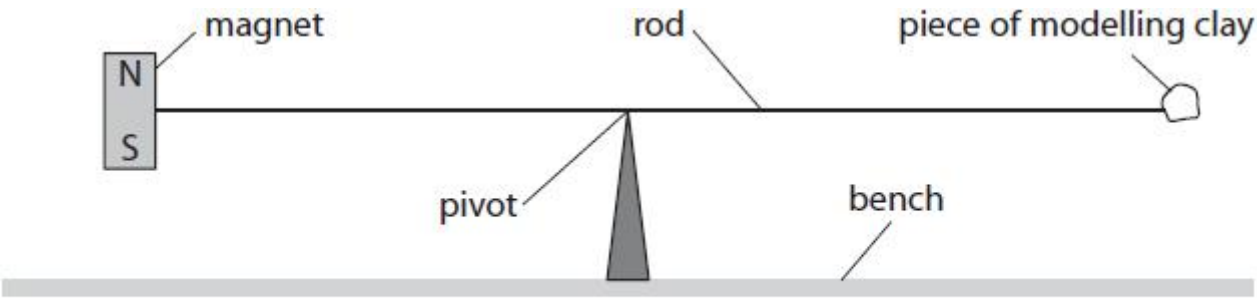


**Figure 14**

The pivot is under the centre of the rod.

A magnet is fixed to one end of the rod.

A piece of modelling clay is fixed to the other end of the rod.

The system is in equilibrium.

The student reverses the direction of the current in the coil.
Describe how the student can bring the system back into equilibrium without making any changes to the magnet.

**(2)**

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**(Total for question = 2 marks)**

**Q10.**

Some forces act at a distance.

One example is the gravitational attraction between the Moon and the Earth.

Describe an example of another type of force acting at a distance, where the force is **not** gravitational.

**(2)**

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**(Total for question = 2 marks)**

**Mark Scheme**

Q1.

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **(i)** | recall of moment = force x distance<br>(1)<br><br>(moment of force from person =) 600 x 0.5<br>**and**<br>(moment of weight of rock =) 1800 x 0.2<br>(1)<br><br>**moment** of force from person is less than **moment** of weight of rock.<br>(1) | may be implied in a calculation<br><br>300 (Nm)<br><br>**and**<br>360 (Nm)<br><br>independent mark accept reverse argument | **(3)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | An explanation that links<br><br>increase distance between person and pivot/ reduce distance between rock and pivot / increase force from person (1)<br><br>increase the moment of the force from the person / decrease the moment of the weight of the rock (1) | <br><br>use longer lever / hold lever nearer the end / move pivot nearer to rock / get someone to help to push<br><br>value of new distance and calculation of new moment | (2) |

**Q2.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | an explanation linking<br><br>distance from hinge/pivot increased (1)<br><br>(therefore)<br>smaller force needed (to close door) | <br><br>P further from hinge than Q<br><br><br>accept<br>the greater distance gives greater moment for 2 marks | (2) |

**Q3.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **(i)** | calculation of both moments (1)<br><br>moments are equal (size) and opposite direction (1) | 4 x 92 and 16 x 23 seen<br><br>comparison of both moments<br><br>368 (=) 368<br>**or**<br>4 x 92 = 16 x 23<br>**or**<br>$16 = \frac{4 \times 92}{23}$<br>**or**<br>$4 = \frac{16 \times 23}{92}$<br><br>accept calculations in Nm | **(2) AO1, AO2** |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **(ii)** | calculation of moment of ball (1)<br><br>480 (Ncm)<br><br>calculation of total clockwise moment (1)<br><br>848 (Ncm)<br><br>calculation of bicep force needed (1)<br><br>212 (N)<br><br>**OR**<br><br>calculation of moment of ball (1)<br><br>480 (Ncm)<br><br>calculation of additional force from bicep (1)<br><br>120 (N)<br><br>calculation of total bicep force (1)<br><br>212 (N) | <br><br>15 x 32 seen<br><br><br><br>368 + 480 seen<br><br><br><br>848/4 seen<br><br><br><br><br><br>15 x 32 seen<br><br><br><br>480/4 seen<br><br><br><br>120 + 92 seen<br><br>award full marks for correct answer without working.<br><br>accept conversion of cm to m throughout | **(3)**<br>**AO2** |

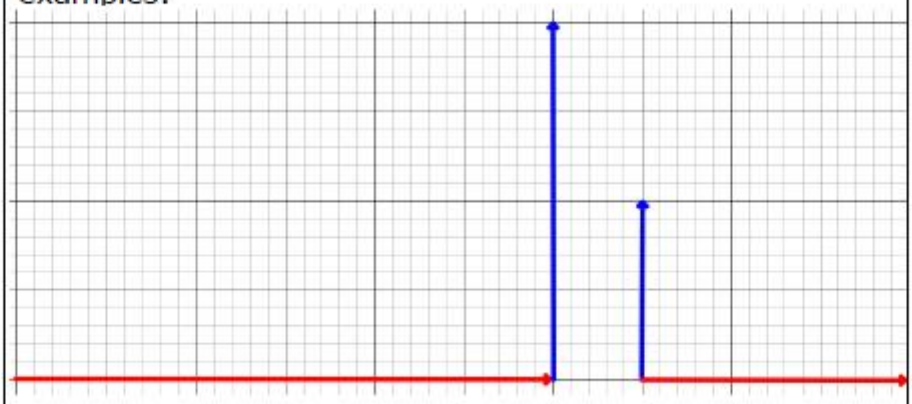
**Q4.**

| Question number | Indicative content | Mark |
|---|---|---|
| | Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme.<br><br>The indicative content below is not prescriptive, and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant.<br><br>A01<br><br>**Horizontal forces**<br>• tension in the string/pull of the string<br>• tension is one of the horizontal forces acting on the wooden block<br>• friction between the table and the wooden block<br>• friction acts in the opposite direction to the tension<br>• friction opposes motion<br>• the force due to friction is equal to the force provided by the tension<br>• the forces are balanced /equal and opposite<br>• no resultant force, so the block moves at a constant (horizontal) velocity | **(6)** |
| | **Vertical forces**<br>• (normal) reaction (force) upwards between the table and the wooden block<br>• contact force<br>• weight of block downwards<br>• the weight (force of gravity) and the (normal) reaction are equal and opposite / balanced<br>• the block does not move up or down<br>• tension caused by the force due to gravity on the weight<br>• vertical forces on the block do not affect horizontal velocity.<br><br>labels on the diagram should be considered when marking<br>candidates do not have to indicate the forces as horizontal and vertical | |

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | • No rewardable material. |
| Level 1 | 1-2 | • Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1)<br>• Presents an explanation with some structure and coherence. (AO1) |
| Level 2 | 3-4 | • Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1)<br>• Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1) |
| Level 3 | 5-6 | • Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1)<br>• Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1) |

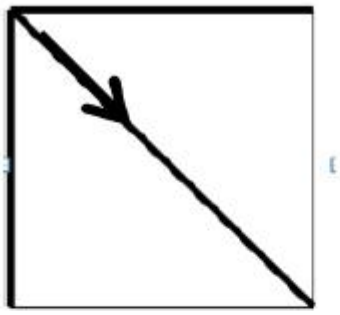
| Level | Mark | Additional Guidance | General additional guidance – the decision within levels<br><br>e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level. |
|---|---|---|---|
| | 0 | No rewardable material. | |
| Level 1 | 1–2 | Additional guidance<br><br>Elements of physics present i.e. isolated knowledge of principles **two** unconnected statements from any section | Possible candidate responses<br><br>contact force<br><br>weight pulls block<br><br>friction<br><br>tension in string<br><br>weight acts downward<br><br>reaction |
| Level 2 | 3–4 | Additional guidance<br><br>Some knowledge of principles with a logical connection made in one section and statement from the other section<br><br>OR<br><br>Detailed knowledge of principles with logical connections made in one section | Possible candidate responses<br><br>There is friction between the table and the block and this opposes motion<br><br>weight of block acts downwards |
| Level 3 | 5–6 | Additional guidance<br><br>Detailed knowledge of principles with logical connections made in one section and statement from the other section | Possible candidate responses<br><br>Friction and tension are equal and opposite there is no resultant /no acceleration<br><br>The weight (of the block) and the(normal) reaction are equal and opposite |

**Q5.**

| Question number | Answer | Mark |
|---|---|---|
| (i) | • two vector arrows at rights angles representing the forces (1)<br>• two vector arrows in proportion (1)<br>examples:<br>  | (2) |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | • drawing shows a completed triangle or parallelogram (1)<br> <br>• Evaluation 3.6 N (1) | ±0.2 N may be calculated using Pythagoras theorem | (2) |

**Q6.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | Scale drawing<br><br>two lines at right angles (1)<br><br>a correct scaling (for example 10kN equivalent to 1 cm) / a completed square or triangle(1)<br><br>diagonal in correct direction (1)<br><br>28 kN (1) |  <br><br>judge by eye<br><br>accept answers from 25 kN to 30 kN<br><br>accept use of Pythagoras<br><br>award full marks for correct answer without working. | **(4)** |

**Q7.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | (upward) force increases with speed (1)<br><br>relationship is non-linear (1) | allow reverse argument<br><br>changing rate / increases exponentially/ initially no upward force (until 1000 turns per minute) | **(2)** |

**Q8.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | any **three** from | marks can be awarded from a correctly labelled diagram | **(3)**<br>**AO1** |
| | magnetic **fields** interact (1) | magnets are in each other's magnetic **field** | |
| | (force due to) repulsion (between magnets) (1) | repel / push away | |
| | (repulsion) force upwards (on upper magnet) (1) | | |
| | weight / gravitational force (downwards on upper magnet) (1) | accept gravity (acts downwards) | |
| | forces equal size / in equilibrium (1) | forces are balanced | |
| | | ignore references to charge | |

**Q9.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | move the (position of) the (0.050 N) weight (1)<br><br>to the other side of the pivot/3.6 cm from the magnet (1) | adjust mass of modelling clay<br><br>reduce (mass of modelling clay) by taking some away<br><br>add (additional) weight between pivot and magnet scores 2 marks | **(2)**<br>**AO3** |

**Q10.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | An answer that combines the following points of understanding to provide a logical description:<br><br>named force (acting at a distance) (1)<br><br>situation (1) | <br><br><br>e.g. magnetic<br><br>force between two (magnetic) poles | **(2)** |